Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Thomas Dankers	Contact	thomas.dankers@tno.nl
Institute Name	Mobility & Built Environment Unit, Powertrains Department			
Address	Postal: P.O. Box 756, 5700 AT Helmond, The Netherlands			

2. Technology Overview

Title of Proposed Technology	Hybrid Vessel Energy Management and Battery Monitoring for High-Speed Workboats (CTV-class vessels)			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communications		Electrical & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Software	SW Solutions	Batteries	Secondary Batteries
Concept of Proposed Technology	Core concept • Model-based Energy Management System (EMS) and Battery Management System (BMS) toolchain for hybrid workboats (high-speed crew transfer vessels and similar duty cycles). • Physics-based battery and powertrain models combined with real-time capable supervisory control to optimize power split (gensets and battery) while enforcing safety, power, and lifetime constraints. • Embedded state estimation (SOC/SOH) and health diagnostics to enable predictive			

	maintenance, safer operation, and lifetime-aware scheduling of charge and discharge. Target use cases (examples) • Peak shaving and load leveling for propulsion transients, hotel loads, and dynamic positioning. • Mission and route aware control using representative duty cycles (port maneuvers, transit, station keeping). • Data outputs aligned with emerging sustainability and traceability requirements (battery passport readiness where applicable).
Rationale for Joint R&D Project	Why a joint R&D project is needed • Hybrid workboats face highly dynamic power demand profiles and strict reliability and safety requirements. Baseline rule-based control can leave significant efficiency and lifetime potential unrealized. • The Korean SME partner contributes vessel integration know-how, shipboard power system architecture, representative operational profiles, and demonstration capability. • TNO contributes validated modelling, optimization-based control (including MEMS), advanced battery diagnostics, and laboratory validation facilities to reduce technical risk before sea trials.
Key Words	1. Hybrid vessel EMS 2. Battery monitoring and control (SOC/SOH/RUL) 3. Lifetime and degradation-aware control 4. Safety and thermal risk mitigation 5. Digital twin and model-based design

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (6)
R&D Objectives	R&D Objectives (technical) • Co-design and validate a hybrid vessel EMS using TNO's MEMS framework: multi-objective optimization balancing fuel/energy cost, emissions, battery ageing and operational constraints. • Develop a battery digital twin for the vessel battery system (cell-to-pack), enabling robust SOC/SOH estimation and degradation-aware control. • Implement health diagnostics and fault detection relevant for maritime operation (sensor plausibility, thermal events, power capability fade). • Deliver a real-time capable embedded implementation and verification workflow (SIL/HIL), including interface specifications for shipboard integration.			
R&D Target Specifications	R&D Target Specifications (proposed targets, to be refined with partner data) • Target 5–15% reduction in fuel consumption over representative vessel missions versus baseline rule-based control (depending on duty cycle and hybrid architecture). • Target 10% reduction in battery cumulative throughput or equivalent ageing metric via degradation-aware scheduling. • Real-time control execution: ≤ 100 ms cycle time on target embedded hardware, with deterministic fail-safe modes. • SOC estimation error target: ≤ ±3% in normal operating region; SOH tracking reported with confidence bounds. • Validation of safety constraints (thermal limits, isolation monitoring, fault handling) through controlled lab tests and on-vessel verification where possible.			
R&D Plan	R&D Plan (indicative 36-month structure) WP1 Requirements and mission profiling (Months 1–6) • Collect and anonymize representative duty cycles, power profiles, and operational constraints from the Korean SME partner. • Define hybrid architecture variants and baseline control strategies for benchmarking. WP2 Physics-based modelling and digital twin (Months 3–15) • Develop and parameterize electro-thermal and ageing models for the vessel battery system.			

	• Create vessel-level power, propulsion, and auxiliary model including converters and genset dynamics. WP3 Control design (Months 6–24) • Configure MEMS-based supervisory control for vessel operation (power split, charge sustaining, constraint handling). • Integrate state estimation (SOC/SOH) and health diagnostics (fault detection and isolation). WP4 Verification, validation, and safety assessment (Months 12–33) • SIL/HIL validation including fault injection and extreme operational cases. • Laboratory testing of battery modules/packs to calibrate and validate models and algorithms. WP5 Integration and demonstration (Months 24–36) • Deploy controller on target hardware and integrate with shipboard systems (interfaces, logging). • Demonstrate performance on partner platform (dock tests; sea trials where feasible). • Deliver documentation, parameter sets, and an exploitation roadmap for commercialization.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO (research institute) responsibilities • Provide a model-based development toolchain: physics-based modelling, parameter identification, and digital twin development (see refs. [1], [3], [6]). • Design and validate EMS/BMS algorithms (MEMS-based supervisory control, SOC/SOH/RUL estimation, diagnostics) and provide a SIL/HIL verification workflow (refs. [2], [3], [4], [5]). • Execute laboratory testing and validation at the Powertrains Test Centre and Battery Lab in Helmond, including: - Battery cell and module testing, including EIS characterization and ageing test campaigns (ref. [10]). - Climate battery pack testing with bidirectional 750 V / 400 A / 300 kW capability, temperature range -60°C to +150°C, 5%RH to 95%RH, and optional inert nitrogen environment (ref. [11]). - Climatic altitude HD-vehicle testing (up to 4000 m equivalent pressure, -45°C to +55°C) for validation of control strategies under extreme conditions (ref. [11]). • Support system integration guidance (interfaces, calibration, verification protocols) and contribute to joint IP and publications.
Expected R&R of Korean SME Partner	Korean SME partner responsibilities • Provide vessel architecture, integration constraints, and performance requirements. • Provide representative mission profiles and access to demonstration platform for integration testing and, where feasible, sea trials. • Implement shipboard integration (power distribution, EMS/BMS interfacing, HMI/logging) and support regulatory alignment for marine deployment. • Contribute to commercialization planning and exploitation of results in marine product lines.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected Outcomes and Impact • Validated hybrid-vessel energy management and battery monitoring solution ready for integration into a commercial high-speed workboat platform. • Quantified improvements in fuel consumption, operational cost, and battery lifetime through degradation-aware control and optimized mission strategies. • Reusable digital twin and SIL/HIL verification workflow reducing development time for future vessel variants and electrified marine applications. • Pathway to joint IP, publications, and follow-on demonstrators under the TIPA programme.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization